DSA 210 Project Proposal
Project Title:
Impact of Content Consistency on Social Media Growth

Student Name:
Zainab Siddiqui (36082)

Data Source

The primary dataset will be obtained from YouTube by selecting 5–10 content creators within a single focussed segment (e.g., podcasts or educational content) to control for content-type variation. For each creator, I will extract data from their most recent 20–30 videos, ensuring a consistent and comparable timeframe across channels. The dataset will include publicly available parameters such as upload date, video duration, number of views, likes, and comments.

To improve the dataset, I will construct additional behavioral variables generated from raw timestamps, including posting intervals and upload frequency patterns.

Data Collection Method

The data will be collected through a combination of manual extraction and automated tools such as the YouTube Data API or browser-based scraping utilities. The collection process will include systematically recording video-level information and categorizing it for each creator.

Following data collection, I will perform preprocessing steps including timestamp standardization, removal of outliers (e.g., unusually viral videos), and normalization of engagement metrics. I will then assess features such as time gaps between consecutive uploads, rolling averages of engagement, and consistency indices (e.g., variance in posting intervals).

Data Characteristics

The final dataset is expected to contain approximately 150–300 observations, where each observation corresponds to an individual video. The dataset will include both raw variables (views, likes, comments, upload time, duration) and generated features (upload interval, average posting frequency, engagement rate).

The data will have a **panel time-series structure**, grouped by content creator and ordered chronologically. This structure enables analysis of both cross-sectional differences between creators and time dependant trends within each channel. The dataset will allow for investigation of relationships between posting consistency and engagement outcomes, as well as variability in performance over time.